

Project Report: Production ML Platform with Monitoring - ML / AI Engineer

Written by Gagan Pasupuleti

Family:	ML / AI Engineer
Level:	Advanced
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

Batch scoring pipeline with drift detection, model registry, and automated retraining trigger.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Advanced]

Chapter focus: Project Overview** *(Level: Advanced)

Production ML Platform with Monitoring** is a advanced-level end-to-end project for ML / AI Engineer. Duration: 5-6 weeks. Batch scoring pipeline with drift detection, model registry, and automated retraining trigger. Tools: MLflow, Evidently, Airflow, scikit-learn.

Code Reference:

```
Objective: Scheduled batch scoring job
Objective: Evidently drift detection report
Objective: Model registry with versioning
Objective: Retrain trigger when drift exceeds threshold
```

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

ML / AI Engineer portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a ML / AI Engineer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: Batch [Advanced]

Chapter focus: Phase: Batch** *(Level: Advanced)

Scoring pipeline. Complete these tasks: Feature store read; Batch predict. Deliverable for this phase: Daily scores table.

Code Reference:

- [] Feature store read
- [] Batch predict

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Batch is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

ML / AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Daily scores table to the teacher.

Chapter 3: Phase: Drift [Advanced]

Chapter focus: Phase: Drift** *(Level: Advanced)

Evidently reports. Complete these tasks: Data drift; Target drift. Deliverable for this phase: Weekly drift report.

Code Reference:

- [] Data drift
- [] Target drift

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Drift is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

ML / AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Weekly drift report to the teacher.

Chapter 4: Phase: Registry [Advanced]**Chapter focus: Phase: Registry** *(Level: Advanced)**

MLflow model registry. Complete these tasks: Staging ? Production; Rollback. Deliverable for this phase: Registry screenshot.

Code Reference:

- ☐ Staging ? Production
- ☐ Rollback

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase Registry is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

ML / AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Registry screenshot to the teacher.

Chapter 5: Phase: Retrain [Advanced]**Chapter focus: Phase: Retrain** *(Level: Advanced)**

Automated trigger. Complete these tasks: Airflow DAG; Champion/challenger. Deliverable for this phase: Retrain run log.

Code Reference:

- ☐ Airflow DAG
- ☐ Champion/challenger

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Retrain is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

ML / AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Retrain run log to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Advanced]

Chapter focus: Final Deliverables and Review *(Level: Advanced)**

Submit all final deliverables: Batch pipeline; Drift reports; Registry setup; Ops runbook. Skills demonstrated: MLOps, Drift detection, Automation. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

Deliverable: Batch pipeline
Deliverable: Drift reports
Deliverable: Registry setup
Deliverable: Ops runbook

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide**What it actually is**

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits Batch pipeline and passes the rubric.